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Forty-third International Conference on Machine Learning

ICML 2026

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We propose SpecGrad, a spec-driven RF inverse design method that unrolls physics-consistent refi

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Analog circuit design automation faces a fundamental challenge known as "Topology-Parameter Entanglement," where physical functions emerge from the non-linear coupling between discrete topologies and continuous parameters. Existing data-driven approaches primarily rely on "Distribution Fitting" to imitate expert patterns. However, they often treat physical specifications merely as post-hoc validation metrics, failing to align the generative process directly with the ultimate physical performance. In this paper, we propose SpecGrad, an end-to-end generative framework for cascaded RF networks that facilitates a paradigm shift from passive imitation to physical attribution. Formulated as a Spec-Driven Reverse Guidance problem via bilevel optimization, SpecGrad explicitly embeds the inner-loop parameter refinement dynamics into the outer-loop generative objective. Crucially, to overcome the gradient blockage caused by discrete topological decisions, we introduce a Physics-Consistent Mixed Gradient Estimation strategy characterized by a "Hard-Execution / Soft-Attribution" mechanism. "By leveraging the linear combinability of Transfer Matrices

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